

ALEJANDRO ITO ARAMENDIA

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EDUCATION

Tohoku University, Japan – *BEng Mechanical and Aerospace Engineering* 2021 – 2025

- GPA: 3.35
- Relevant Coursework: Finite Element Method Analysis (Python), Pygame (Python)

King's College London Maths School, UK – *A-Levels* 2019 – 2021

- A*AAA in Maths, Further Maths, Physics and Computer Science (AS)
- Virtual Sprint Business Idea Competition Winner
- Dynamics of Swing Pumping - King Certificate Project

EXPERIENCE

JIGE Internship Business Competition Top 3 – *Kansai University* Sep 2024 – Nov 2024

- Led and guided a team through the development of a comprehensive business expansion strategy, including market research, risk mitigation, entry planning and pitch presentations.
- Collaborated with team members to evaluate market entry scenarios, ensuring strategic alignment with long-term business objectives and operational feasibility.

Research Assistant – *Ishii Lab, Tohoku University* Oct 2023 – Sep 2024

- Built a MATLAB GUI to automate complex vector fluid flow analysis, increasing data processing efficiency for research teams and enhancing multi-dataset analysis.
- Analysed the propagation speed and flow direction of tissue regions by using a combination of regression and bilinear interpolation techniques.

International PBL Workshop 2023 – *National Taipei University of Technology* Aug 2023 – Aug 2023

- Built an autonomous Arduino robot in C++ capable of recognising coloured balls and delivering them to designated locations.
- Took the lead in team presentations, clearly communicating project goals and updates to a panel of judges and 100+ participants.

PROJECTS

Finance Tracker

- Developed a fully serverless REST API using Node.js, AWS Lambda, and API Gateway, implementing secure JWT-based authentication and CRUD operations for finance data.
- Integrated a comprehensive CI/CD pipeline (GitHub Actions) with automated unit tests (Jest), ensuring high code quality and rapid, reliable deployments to AWS.

SRS Japanese Flashcard Website

- Developed an interactive spaced repetition system (SRS) to support users learning kanji, enhancing study efficiency.
- Built secure user authentication and data storage with Redis, PostgreSQL, and bcrypt, ensuring safe handling of user flashcard progress.

Chess Game

- Built a fully functional chess game using JavaScript, HTML, and CSS with a focus on dynamic move validation and user experience.
- Enhanced user experience by displaying legal moves with red dot indicators, a difficult task for such a complex game, allowing smooth and interactive gameplay flow.

Technical Writer on Machine Learning and Algorithms (2000+ Likes)

- **Articles on ML:** Transformers, CNN, DCNN, U-Net, U-Net++, DenseNet, L1 and L2 Regularization
- **Articles on DSA:** Dijkstra, Quick Sort, Hash Tables, Merge Sort, Time Complexity, Bubble Sort

SKILLS

Programming – JavaScript (ES6+), Node.js, Express, SQL (PostgreSQL, MongoDB), AWS (Lambda, API Gateway, S3), Git, CI/CD (GitHub Actions), Jest (Unit Testing), Postman, Python, Pandas and MATLAB

Languages – English (Native), Spanish and Japanese (Limited Professional Proficiency)